

QUESTIONS 20 / 20 completed

Arithmetic

ARITHMETIC

Write a Python program to perform addition, subtraction, multiplication, and division of two numbers.

PROGRAM EDITOR

Python

Run

Save

```
1 a = float(input("Enter first number: "))
2 b = float(input("Enter second number: "))
3
4 print("\nResults:")
5 print("Addition (a + b) =", a + b)
6 print("Subtraction (a - b) =", a - b)
7 print("Multiplication (a * b) =", a * b)
8
9 if b != 0:
10     print("Division (a / b) =", a / b)
11 else:
12     print("Division (a / b) = Undefined (cannot divide by zero)")
```

INPUT

10
5

OUTPUT

```
Enter first number: Enter second number:
Results:
Addition (a + b) = 15.0
Subtraction (a - b) = 5.0
Multiplication (a * b) = 50.0
Division (a / b) = 2.0
```

QUESTIONS 20 / 20 completed

Modulus ▼

MODULUS

Write a Python program to check if a number is even or odd using the modulus operator.

PROGRAM EDITOR

Python ▼

Run

Save

```
1 n = int(input("Enter a number: "))
2
3 if n % 2 == 0:
4     print("Even")
5 else:
6     print("Odd")
```

OUTPUT

```
Enter a number: Even
```

INPUT

8

QUESTIONS 20 / 20 completed

Comparison

COMPARISON

Write a Python program to check if a number is positive, negative, or zero using the comparison operator.

PROGRAM EDITOR

Python

Run

Save

```
1 num = float(input("Enter a number: "))
2
3 if num > 0:
4     print("Positive")
5 elif num < 0:
6     print("Negative")
7 else:
8     print("Zero")
```

INPUT

5

OUTPUT

```
Enter a number: Positive
```

QUESTIONS 20 / 20 completed

Assignment

ASSIGNMENT

Write a Python program to swap two variables using the assignment operator.

PROGRAM EDITOR

Python

Run

Save

```
1 a = int(input("Enter first number: "))
2 b = int(input("Enter second number: "))
3
4 a = a + b
5 b = a - b
6 a = a - b
7
8 print("After swapping:")
9 print("a =", a)
10 print("b =", b)
```

INPUT

10
20

OUTPUT

```
Enter first number: Enter second number: After swapping:
a = 20
b = 10
```

QUESTIONS 20 / 20 completed

Concatenation



CONCATENATION

Write a Python program to concatenate two strings using the concatenation operator.

PROGRAM EDITOR

Python



Run

Save

```
1 s1 = input("Enter first string: ")
2 s2 = input("Enter second string: ")
3
4 result = s1 + s2
5 print("Concatenated string:", result)
```

INPUT

Hello
World

OUTPUT

```
Enter first string: Enter second string: Concatenated string: HelloWorld
```

QUESTIONS 20 / 20 completed

Modulus

MODULUS

Write a Python program to find the remainder of a division using the modulus operator.

PROGRAM EDITOR

Python

Run

Save

```
1 a = int(input("Enter dividend: "))
2 b = int(input("Enter divisor: "))
3
4 remainder = a % b
5 print("Remainder =", remainder)
```

INPUT

```
10
3
```

OUTPUT

```
Enter dividend: Enter divisor: Remainder = 1
```

QUESTIONS 20 / 20 completed

Logical operations ▾

**LOGICAL
OPERATIONS**

Write a Python program
to perform bitwise AND,
OR, and XOR operations
on two integers.

PROGRAM EDITOR

Python ▾

Run

Save

```
1 a = int(input())
2 b = int(input())
3 print(a & b, a | b, a ^ b)
```

INPUT

5
3

OUTPUT

```
1 7 6
```

QUESTIONS 20 / 20 completed

bitwise operator ▾

BITWISE OPERATOR

Write a Python program to check if a given number is a power of two using the bitwise operator.

PROGRAM EDITOR

Python ▾

Run

Save

```
1 n = int(input())
2
3 if n > 0 and (n & (n - 1)) == 0:
4     print("True")
5 else:
6     print("False")
```

OUTPUT

True

INPUT

1

QUESTIONS 20 / 20 completed

shift operators ▾

SHIFT OPERATORS

Write a Python program to perform left and right shift operations on an integer.

PROGRAM EDITOR

Python ▾

Run

Save

```
1 n = int(input())
2 s = int(input())
3 print(n << s)
4 print(n >> s)
```

OUTPUT

```
32
2
```

INPUT

```
8
2
```

QUESTIONS 20 / 20 completed

ternary operator

TERNARY OPERATOR

Write a Python program to find the maximum of two numbers using the ternary operator.

PROGRAM EDITOR

Python

Run

Save

```
1 a=float(input()); b=float(input())
2 print("Maximum:", a if a>b else b)
```

INPUT

```
10
20
```

OUTPUT

```
Maximum: 20.0
```

QUESTIONS 20 / 20 completed

area of a rect

AREA OF A RECT

Write a Python program to calculate the area of a rectangle.

PROGRAM EDITOR

Python

Run

Save

```
1 l = float(input("Enter length: "))
2 w = float(input("Enter width: "))
3 print(l*w)
```

INPUT

5
4

OUTPUT

```
Enter length: Enter width: 20.0
```

QUESTIONS 20 / 20 completed

area of a triangle ▾

AREA OF A TRIANGLE

Write a Python program to calculate the area of a triangle.

PROGRAM EDITOR

Python ▾

Run

Save

```
1 b=float(input("Enter base: "))
2 h=float(input("Enter height: "))
3 print("Area =", 0.5*b*h)
```

INPUT

10
5

OUTPUT

```
Enter base: Enter height: Area = 25.0
```

QUESTIONS 20 / 20 completed

circumference ▾

CIRCUMFERENCE

Write a Python program to calculate the circumference of a circle.

PROGRAM EDITOR

Python ▾

Run

Save

```
1 r = float(input("Enter the radius: "))
2 pi = 3.141592653589793
3 circumference = 2 * pi * r
4 print("Circumference =", circumference)
```

INPUT

7

OUTPUT

```
Enter the radius: Circumference = 43.982297150257104
```

QUESTIONS 20 / 20 completed

palindrome or not ▾

PALINDROME OR NOT

Write a Python program to check if a string is a palindrome or not.

PROGRAM EDITOR

Python ▾

Run

Save

```
1 s = input().replace(" ", "").lower()
2 print("Palindrome" if s == s[::-1] else "Not Palindrome")
```

INPUT

racecar

OUTPUT

Palindrome

QUESTIONS 20 / 20 completed

factorial



FACTORIAL

Write a Python program to find the factorial of a number.

PROGRAM EDITOR

Python



Run

Save

```
1 n=int(input())
2 f=1
3 for i in range(2,n+1):
4     f*=i
5 print(f)
```

INPUT

5

OUTPUT

120



QUESTIONS 20 / 20 completed

LCM

LCM

Write a Python program to find the LCM of two numbers.

PROGRAM EDITOR

Python

Run

Save

```
1 import math
2 a=int(input())
3 b=int(input())
4 print(abs(a*b)//math.gcd(a,b))
```

INPUT

12
18

OUTPUT

36

QUESTIONS 20 / 20 completed

GCD

GCD

Write a Python program to find the GCD of two numbers.

PROGRAM EDITOR

Python

Run

Save

```
1 def gcd(a, b):  
2     while b != 0:  
3         a, b = b, a % b  
4     return abs(a)  
5  
6 a = int(input())  
7 b = int(input())  
8 print(gcd(a, b))
```

INPUT

24
18

OUTPUT

6

QUESTIONS 20 / 20 completed

Reverse a string ▼

REVERSE A STRING

Write a Python program to reverse a string.

PROGRAM EDITOR

Python ▼

Run

Save

```
1 s = input("Enter a string: ")
2 print(s[::-1])
```

INPUT

Hello

OUTPUT

```
Enter a string: olleH
```

QUESTIONS 20 / 20 completed

Ascending order ▾

**ASCENDING
ORDER**

Write a Python program
to sort a list in ascending
order.

PROGRAM EDITOR

Python ▾

Run

Save

```
1 numbers = list(map(int, input("Enter numbers separated by space: ").split()))
2 numbers.sort()
3 print("Sorted in ascending order:", numbers)
```

OUTPUT

```
Enter numbers separated by space: Sorted in ascending order: [1, 2, 3, 5, 9]
```

INPUT

5 2 9 1 3

QUESTIONS 20 / 20 completed

Remove duplicate ▾

REMOVE DUPLICATE

Write a Python program
to remove duplicate
elements from a list.

PROGRAM EDITOR

Python ▾

Run

Save

```
1 a=[1,2,2,3,1,4]
2 res=[]
3 for x in a:
4     if x not in res: res.append(x)
5 print(res)
```

INPUT

```
[1, 2, 2, 3, 1, 4]
```

OUTPUT

```
[1, 2, 3, 4]
```